

The Path to Automated Narrative Processing: Narrativity, Uncertainty, and Slow Feature Analysis

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Abstract

How does narrative figure into the drive towards artificial intelligences that bear comparison to human cognition? There is a robust literature about narrative computation, focusing on the analysis of narrative structures, as well as large language models' ability to generate narratives. While these directions automate two high-level human cognitive activities, the processes do not resemble how narrative structures are generated and processed into abstractions and memories in human thinking. This presentation describes the function of ongoing narrative processing in low-level cognition, followed by consideration of two robust areas of research that might advance our understanding and anticipate the role it might play in artificial intelligence. The presentation begins by considering the problem ongoing narrative processing helps us solve. I will review and update a model of narrativity as an emergent feature in event processing predicated on parsing event boundaries via extant cognitive models, uncertainty reduction, affect, and attributing meaning (?). I then focus on two key, but limited, topics within this field. In cognitive science, new models of decision-making theorize a key role for uncertainty in time segmentation and abstraction in event processing (?). In artificial intelligence, 'slow feature analysis' works to identify event constituents and define event boundaries (?). I will argue for relationships between these processes and emergent narrativity in event processing. Taken together, these perspectives contribute to research directions in cognitive studies, artificial intelligence, and narrative theory around the question of what human-like event processing would look like in AI. Finally, I consider embodiment, processing limitations, and the ethics of this research direction.

Keywords: narrative processing, artificial intelligence, event cognition, AGI, narrative computation

1 Introduction

How does narrative figure into the drive towards artificial intelligences that bear comparison to human cognition?

There is a robust literature about narrative computation, focusing on the analysis of narrative structures. Large language models can be leveraged to generate narratives. While these two directions automate two high-level human cognitive activities, the processes do not resemble how narrative structures are recognized and created in human thinking.

This article describes the function of ongoing narrative processing in human intelligence, followed by consideration of the state of artificial intelligence in approximating that functionality.

2 The Problem Narrative Processing Solves

My previous research advanced the following hypothesis: Ongoing narrative processing helps us solve a fundamental cognitive problem: orienting ourselves towards, framing responses to, and retroactively parsing experience.

Given that hypothesis, today I want to consider a question: if narrative processing is crucial to human cognition, is it a crucial direction for research pushing AI towards human-like AGI?

2.1 Word Search Junkies

This question motivates me in part because of the conversation about AGI we've been having since December 2022, when OpenAI released ChatGPT. After a wave of articles projecting the capabilities of large language models into a future artificial general intelligence, there has been a noticeable series of popular public pushbacks.

Ted Chiang's recent piece in the Atlantic is a good reminder of what large language models are: sentence completion machines. Chiang's consideration of the leap from simulation to actuality – simulating a conversation to actually conversing, for example, Anthropic's anthropomorphic identity document.

Zeynep Tufekci's recent piece in the New York Times is a good reminder of what large language models are: sentence completion machines.

Robert Long, in his SubStack *Experiencing Machines*, characterizes these arguments as LLM-sneering, where doubters of the capabilities of large language models might say something like "You think that LLMs could be conscious? Absurd! Don't you know that LLMs are just predicting the next token? Plus, they make stuff up; their preferences are contradictory; their introspection is unreliable; and they're confused about who they even are!"

On the other side, defenses of LLMs' intelligence might retort "Aha, well... aren't humans just predicting the next token? Plus, humans make stuff up, and humans have contradictory preferences; our introspection is unreliable, and we're really confused about who we are too!"

Long encourages us not to adopt either of these straw man positions, but instead consider both differences in how LLMs and humans are trained and behave, as well as the sources of the kinds of errors they make.

My talk will operate in this spirit – I want to think with you about how AI might be trained to more closely resemble human cognition, with an emphasis on the topic that most concerns us: the relationship between narrative and cognition.

For starters, consider one of Long's observations about how LLMs differ from humans: LLMs "learned language first and would bootstrap any potential experiences from that, whereas humans can have experiences independently of using language to describe them."

I've seen related claims about LLMs' lack of phenomenal, perceptual, proprioceptive, emotional, and other processes we consider human.

It may make sense to ask what it is like to be a bat, but asking a similar question about LLMs seems dubious.

If we are seeking developments in AI that might have increased similarity to human cognition, I believe the 'bootstrapping' problem means it makes sense to look elsewhere.

I believe we should look areas of research in computer vision.

Now, I imagine that some of you have watched videos of Waymo cars endlessly circling roundabouts and Atlanta neighborhoods, so you have good reason for skepticism.

Even so, I think moving vehicles do not ipso facto have the bootstrapping problem.

Also, I think computer vision research will increasingly focus on one of the bases of having experiences: event segmentation.

Event Segmentation Theory (EST) offers a precise account of where event boundaries come from. An event is, at minimum,

“a segment of time at a given location that is perceived by an observer to have a beginning and an end” (?, p. 128)

and boundaries between events are generated dynamically, as a byproduct of prediction:

“One central proposal of the EST model is that event segmentation is a spontaneous outcome of ongoing perception and arises at points where perceptual predictions begin to fail or change. . . The period of epoch between two such consecutive changes current mental models is an event.” (?, p. 135)

If we refer back to the 'LLM-sneerers', we note that their critique centers around the reductive way that LLMs are 'generative' – they predict the next token based on previous tokens.

'LLM-defenders' counter not only that humans also predict, but that LLMs can come from complex forms of context.

But I'd like to point out that this form of prediction is quite different from how prediction functions in ongoing experiences.

“As an event proceeds, predictions are relatively accurate. However, as it ends, prediction error increases, causing a reset of the event model. When such a reset occurs, an event boundary is set.” (?, p. 135)

This kind of prediction requires online processing of ongoing experience. Which brings us back to Waymo and computer vision.

It is imperative that Waymos, other autonomous vehicles, and other embodied AI systems have robust event parsing capabilities.

I want to highlight two areas of research that attempt to create computational models related to event parsing: slow feature analysis and processing uncertainty.

Slow feature analysis (SFA) is a computational technique that identifies slowly varying features in time series data, which can be used to detect event boundaries. SFA has been applied in various domains, including computer vision and robotics, to improve the understanding of dynamic environments.

According to ?, SFA extracts temporal features whose 'slow' organizing principle is consistent with the nature of biological vision (?) " Types of SFA.

What is Coglinks?

If online SFA in computer vision can continue to be refined to incorporate the strengths of probabilistic, multi-modal, and other forms of SFA, it might be employed in conjunction with salience systems to provide ongoing, continuously updating prediction failure processing and equilibrium-disequilibrium-tracking. These elements generate emergent event recognition.

In this context of ongoing emerging event cognition, computational approaches to uncertainty like Coglinks can be developed to transition from modeling uncertainty about well-defined choices to modeling the emergence of choices from the fuzzy, uncertain domain of predictive processing, of the navigation of choices marked by uncertainty, including ill-defined problems, and of learning from online decision navigation, taking the form of abstracting from event experiences.

That's the gist of what I wanted to explore today. Please note that this is a sketch, and an incomplete one at that.

Two topics about what needs to be thought through next jump to mind: first, what is the nature of the valuation systems that support emergent event processing in human cognition, and how might they be modeled computationally? In Narrative in Cognition, I ascribed significance to the emotional processes that support event cognition: emotional systems support ongoing, dynamic valuations of salience and valence. In the function of these systems in modulating the equilibrium-disequilibrium and prediction failure patterns of event processing I see weak narrativity.

How can ongoing valuation systems be designed so that event cognition leverages valuation dynamism? That is, how can events be perceptually emergent?

This leads to the second topic: perceptually emergent events must depend, in part, on a dynamic, systemic attention architecture. a. attention and prediction failure b. attention and the awareness model

These are topics for future consideration. They lead, subsequently, to questions about how to model human-like learning and abstraction. How are event segments shaped by memory of previous events, and how do salient, valent events get abstracted into general knowledge, as human cognition does with its ongoing processing of episodic into semantic memory?

2.2 Event parsing: the challenges of recognition;

Narrativity and Cognition model of event parsing. role of attention NC theory of how attention is modulated division between 'functional understanding and phenomenological understanding' (DHSI keynote) is reductive – as we develop systems that approach intelligence, we must question whether 'phenomenological understanding' is the root of what we mean by 'intelligence', or whether it is simply a necessary partner for humans, a fundamental part of the processing that leads to intelligence. that distinction sounds small, but if AI is to be pushed towards 'intelligence', it is important, because perhaps the links between how intelligence develops in humans and how it develops in AI are How event parsing is subsumed in larger time series think about the Waymo endless circling in terms of this now, we COULD just solve this with a 'Do' loop but that differs significantly from human intelligence, and the larger motivations, cares, and needs

that drive, curtail, and otherwise shape our behaviors what is the nature of these time series?

2.3 Orienting Towards Experience

2.4 Framing Responses

2.5 Retroactive Parsing

3 Human Event Processing

4 AI Event Processing: State of the Art

4.1 Narrative Analysis Systems

4.2 Narrative Generation

4.3 Gaps and Limitations

5 A Research Roadmap

5.1 Direction 1

5.2 Direction 2

5.3 Contributions to AGI

6 Ethics and Open Questions

6.1 Is Embodiment Essential?

Is embodiment an essential precondition of narrative processing?

6.2 Limitation or Feature?

Is the “functionality” of narrative processing actually a limitation imposed by processing capacity, and might artificial systems ultimately transcend that limitation?

7 Conclusion

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